



Predicting the strength of next-day negative emotion states in body dysmorphic disorder using passive smartphone data: An intensive longitudinal assessment study

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ABSTRACT

Body dysmorphic disorder (BDD) is a debilitating and common psychiatric illness associated with high rates of suicide and substance use disorders. Negative emotions – particularly shame and anxiety – are elevated in BDD and correlate with suicide risk and substance use. It is critical to have reliable and valid tools to assess negative emotions in BDD. Retrospective self-reports are subject to recall biases, average one's experiences over broad time frames, and are burdensome to complete. Alternatively, sensor-based digital phenotyping has potential to yield low-burden emotion assessment within acute time frames. This study aimed to use smartphone sensor data (GPS, accelerometer, collected over 3 months) to predict next-day peak shame, anxiety, and general negative emotion states (collected via 28 days of ecological momentary assessment) in 83 adults with BDD. We tested cumulative link mixed models [CLMM] and random forest [RF] models. RFs outperformed CLMMs across prediction performance metrics and had overall prediction accuracies (i.e., proportion of predicted scores that exactly matched actual scores, out of total predictions) of 42.1–50.0 %, versus 10.9–20.2 % for CLMMs. Binary predictive performance at high levels of negative emotion was moderate. Developing unobtrusive methods for predicting shame, anxiety, and general negative emotion states over acute time frames using smartphone sensor data can enable just-in-time intervention opportunities, as a future step to reduce risk for suicide and substance use in BDD. Models might be strengthened with larger samples, data collected over longer time frames, and incorporation of wearable-based physiological data.

Trial Registration: [ClinicalTrials.gov](https://clinicaltrials.gov) Identifier: NCT04254575.

1. Introduction

Body dysmorphic disorder (BDD) is a chronic and debilitating psychiatric illness that affects 1.7–2.4 % of the population (Buhlmann et al., 2010; Koran et al., 2008; Rief et al., 2006; Schieber et al., 2015). BDD involves a persistent worry about physical appearance and time-consuming, maladaptive rituals to manage appearance concerns (American Psychiatric Association, 2013). Negative emotions – particularly shame and anxiety – are central to BDD (Weingarden and

Renshaw, 2016). Shame is felt when judging oneself in a stable and harshly critical manner (e.g., “I’m defective”) and generates harmful behavioral responses including social withdrawal and isolation (Tangney and Dearing, 2002). Anxiety involves worry about a distal threat, physiological arousal, and safety behaviors (Encyclopedia of Psychology, n.d.) including rituals and avoidance. Shame and anxiety are elevated in people with BDD compared to clinical and non-psychiatric samples (Weingarden et al., 2018; Weingarden et al., 2016) and are linked with risk sequelae including suicidal thoughts

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(Weingarden et al., 2018; Weingarden et al., 2017), functional impairment (Weingarden et al., 2016; Weingarden et al., 2017), and depression symptoms (Weingarden et al., 2016).

Given the central role of negative emotions in BDD and their relationship with risk outcomes, it is important to develop valid, time sensitive, and user-friendly tools for their assessment. At present, the field relies on retrospective self-report measures. Retrospective measures are subject to recall biases (Ebner-Priemer and Trull, 2009) such as remembering only the most emotionally salient event when completing a questionnaire asking about the past week, likely missing variability and nuance in affective experiences. Retrospective measures that evaluate broad time frames also cannot tell us about one's current emotional experience, which is particularly important given that emotions are transient and have an acute impact on risk. Finally, self-report measures result in user burden each time new information is needed, lowering the likelihood that clinicians assign these measures or that patients complete them (Wilhelm et al., 2020). Low burden and time-sensitive emotion assessment would enable short-term monitoring and delivery of just-in-time interventions that support people in responding adaptively to negative emotions, to reduce harmful down-stream sequelae.

One promising path is to develop such tools using smartphone-based digital phenotyping; that is, quantifying acute emotion outcomes using passively-collected smartphone data during daily life (Onnela and Rauch, 2016). An estimated 90 % of US adults own a smartphone (Pew Research Center, 2024), making algorithms based on smartphone data deployable without purchasing or wearing additional devices. A wide array of data can be securely collected from smartphones without user burden, and raw sensor data such as GPS coordinates can be distilled into interpretable and validated summary statistics (e.g., distance traveled from home per day) that may map onto behavioral features of negative emotions such as avoidance and social withdrawal.

Research that tests associations between passive smartphone data and negative emotions or their behavioral features is early. Passive smartphone data, including GPS-derived daily home time (Stamatis et al., 2024; Chow et al., 2017) (a potential proxy for avoidance), regularity in circadian movements (Stamatis et al., 2024), and variability in location and mobility features (Stamatis et al., 2024), as well as text frequency and call duration (Cao et al., 2020) (potential proxies for social withdrawal) correlate with measures of mood and anxiety in adolescent, community, undergraduate, and depressed samples (Stamatis et al., 2024; Chow et al., 2017; Cao et al., 2020; Jacobson et al., 2020; Jacobson and Chung, 2020). While prior studies predicted broad constructs of negative mood or anxiety, a recent trial focused on predicting specific emotion states (anger, sadness, nervousness) from sensor data (Ren et al., 2023). In 22 adolescents with depressive symptoms, Ren and colleagues (2022) found that elevated anger was predicted most accurately, and personalized ensemble machine learning models outperformed linear models. Only one pilot trial ($N = 10$) has used sensors to predict psychological phenomena in BDD, finding that GPS-derived time at home correlated with BDD severity ($r = 0.49$) and response to smartphone cognitive behavioral therapy for BDD (Weingarden et al., 2020). Altogether, most prior digital phenotyping research was conducted in convenience or depression samples (Sequeira et al., 2020). Limited work has focused on predicting specific emotions, although there is clinical value in differentially predicting specific (vs. more general) harmful emotions. Finally, beyond the pilot trial noted above, no studies have used digital phenotyping in BDD.

We sought to develop low burden, time-sensitive, and ecologically valid measures of negative emotions in 83 adults with BDD using smartphone sensor data. Outcomes included maximum next-day shame, anxiety, and general negative affect, collected via ecological momentary assessment (EMA). We evaluated linear mixed models (cumulative link mixed model [CLMM]) and non-linear machine learning models (random forest [RF]). We examined feature importance to inform which data streams and summary statistics are most valuable to collect and compute for emotion prediction, and we evaluated model performance,

to assess how useful models might be for future just-in-time interventions. Drawing from prior work (Ren et al., 2023), we expected to find stronger model performance using RF compared to CLMM. We also expected prediction accuracy to be highest for models of general negative affect, compared to models of specific emotions (i.e., shame, anxiety).

2. Methods

2.1. Participants

Eligible participants were U.S.-based adults aged 18+ with a primary psychiatric diagnosis of BDD based on the Mini International Neuropsychiatric Interview (MINI 7.02 (Sheehan et al., 1998)), who were experiencing at least one of the following: (a) thoughts of suicide within the past month, (b) alcohol use within the past 2 weeks, or (c) marijuana use within the past 2 weeks. BDD symptoms were at least moderate in severity, determined by a score ≥ 24 on the Yale-Brown Obsessive Compulsive Scale Modified for BDD (BDD-YBOCS (Phillips et al., 1997)). Participants were required to own an Android or iOS smartphone, have Wi-Fi-enabled Internet access, and own a computer or tablet to complete study procedures. Participants were not eligible if they were experiencing current manic or hypomanic symptoms, current active suicidal thoughts with intent, or BDD severity that warranted higher than outpatient care.

2.2. Procedures

Procedures were approved by the Massachusetts General Hospital Institutional Review Board. This was an approximately 3-month naturalistic assessment study in which multiple data streams were collected. For the present study, we used data from clinician-assessed diagnostic and clinical information (baseline), self-reported demographics (baseline), EMA of negative emotion states (two 2-week periods at baseline and midpoint, respectively), and passive smartphone data (GPS, accelerometer) collected continuously.

Participants were recruited nationally through referrals from colleagues or within our hospital clinic and research program, professional organizations, posts and advertisements on social media-based (i.e., Facebook, Instagram, Reddit) support groups and mental health discussion boards.

2.2.1. Screening and baseline virtual appointment with clinician

Those who appeared eligible following a pre-screening survey were scheduled for a consent, screening, and baseline virtual appointment with a study assessor. Assessors included a licensed psychologist with specialized training in BDD and a Masters-level predoctoral fellow in psychology within the same BDD specialty program. Both assessors completed training, and assessments were audiotaped to verify reliability through ratings (15 %, randomly selected) by a separate trained, blinded assessor (intraclass correlation [2,1] for BDD-YBOCS = 0.93, 100 % agreement on BDD diagnosis based on MINI).

After logging into a HIPAA-compliant video conferencing platform, participants completed informed consent. The assessor administered diagnostic and clinical assessments (see Measures) to evaluate eligibility criteria and characterize the sample. The research coordinator helped eligible participants download two study apps, Metricwire (Metricwire, 2023) and Beiwe (Onnela et al., 2021), for collection of EMA and raw passive smartphone data, respectively. Finally, participants were directed to complete baseline self-report measures of demographic, clinical, and psychosocial domains via REDCap (Harris et al., 2009; Harris et al., 2019).

2.2.2. Ecological momentary assessments (EMA)

Participants completed two 2-week waves of EMA (28 days total), at baseline and midpoint respectively, on the Metricwire app. We divided

EMA into two waves to reduce burden and increase adherence, and to capture greater variability of experiences of interest by spreading out the time frame covered. EMA included three questionnaires capturing experiences “right now,” sent randomly between 10:00 am–1:30 pm, 1:45 pm–5:15 pm, and 5:30 pm–9:00 pm; questionnaires were available to complete for 30 min once sent.

2.2.3. Smartphone sensor data

Using the Beiwe app, we collected research-grade raw GPS and accelerometer data from participants' iOS and Android smartphones continuously during the 3-month study period. Data were encrypted while buffered on the phone awaiting Wi-Fi upload, and data were re-encrypted on the server once moved over. Raw sensor data (e.g., GPS coordinates over time) were converted into daily summary statistics (e.g., time spent at home per day) using the Forest library (Forest, 2022) created and validated by the Harvard School of Public Health Onnela Lab (see Measures).

2.3. Measures

2.3.1. Clinician-administered measures

A study assessor administered the MINI 7.02 (Sheehan et al., 1998) at screening/baseline to evaluate eligibility criteria and characterize the sample. The MINI is a brief semi-structured diagnostic tool that is well-validated and correlates strongly with the Structured Clinical Interview for Diagnostic and Statistical Manual (Sheehan et al., 1997). The BDD-YBOCS (Phillips et al., 1997) was administered at screening/baseline to evaluate BDD severity to establish eligibility and characterize the sample. The BDD-YBOCS is the gold-standard clinician-administered semi-structured assessment of past week BDD severity. Scores range from 0 to 48, with higher scores indicating more severe BDD symptoms. The Brown Assessment of Beliefs Scale (BABS) (Eisen et al., 1998) was administered at screening/baseline to evaluate BDD-related insight, to characterize the sample. The BABS ranges from 0 to 24, with higher scores indicating poorer insight (i.e., a more fixed belief that appearance concerns reflect a true defect). The Columbia-Suicide Severity Rating Scale (C-SSRS) (Posner et al., 2011) was administered to assess eligibility criteria and characterize suicide risk within the sample.

2.3.2. Self-report questionnaires

At baseline, participants completed a demographics questionnaire, which included the variables shown in Table 1, in addition to sex at birth, sexual orientation, religious affiliation, living situation, and sources of economic support.

2.3.3. Ecological momentary assessment measures

Current negative emotions were measured with items from the Positive and Negative Affect Scale–X (PANAS-X (The PANAS-X: Manual for the Positive and Negative Affect Schedule - Expanded Form - University of Iowa, n.d.; Watson et al., 1988)). The PANAS-X is a well-validated Likert measure of current emotion intensity, ranging from 1 (very slightly or not at all) to 5 (extremely); The PANAS-X reliably assesses between-person differences and within-person change (Watson et al., 1988; Haney et al., 2023). Shame and anxiety were measured with “ashamed” and “anxious” (added for this study) items, respectively, and general negative emotion was measured by averaging all of a participant's negative emotion ratings (i.e., ashamed, anxious, guilty, disgusted with self, sad, angry, agitated) for a given EMA prompt. To match the timing of our independent variables (i.e., daily-level smartphone sensor summary statistics), for each day we used participant's maximum shame, anxiety, and general negative emotion EMA rating. Maximum scores were selected because the *most intense* emotions experienced daily may be most clinically salient, when considering risk monitoring.

2.3.4. Smartphone sensor data

Using the Beiwe app and platform (Onnela et al., 2021), we collected

Table 1

Baseline demographic and clinical characteristics of sample.

	n (%)
Race	
White	62 (74.70)
Black or African American	6 (7.23)
Asian	5 (6.02)
More than one race	8 (9.64)
Another race, not specified	2 (2.41)
Ethnicity	
Hispanic or Latino	13 (15.66)
Gender	
Female	62 (74.70)
Male	17 (20.48)
Genderqueer / nonbinary	2 (2.41)
Unsure / another identity	2 (2.41)
Marital status	
Single / never married	56 (67.47)
Married	13 (15.66)
Living with partner	10 (12.05)
Divorced / separated	4 (4.82)
Employment status^a	
Working full-time	31 (37.35)
Working part-time	22 (26.51)
Student	17 (20.48)
Unemployed	19 (22.89)
Homemaker / stay-at-home parent	6 (6.17)
Education	
High school diploma/GED or less	11 (13.25)
Some college / 2-year degree or certificate	29 (34.94)
College	26 (31.33)
Some post-graduate / professional	3 (3.61)
Post-graduate / professional degree	14 (16.87)
Current psychiatric comorbidities^{a, b}	
At least 1 comorbid diagnosis	83 (100)
Major depressive disorder	30 (36.14)
Panic disorder	11 (13.25)
Agoraphobia	13 (15.66)
Social anxiety disorder	21 (25.30)
Obsessive compulsive disorder	18 (21.69)
Posttraumatic stress disorder	9 (10.84)
Alcohol use disorder	33 (39.76)
Substance use disorder	33 (39.76)
Generalized anxiety disorder	14 (16.87)
Taking psychotropic medication(s) at baseline	49 (59.04)

^a Some subjects reported more than one employment status or comorbid diagnosis.

^b Only lists disorders which have >10 % prevalence.

raw GPS and accelerometer data, sampled intermittently for operational reasons (e.g., preserving battery life). GPS was sampled for 90 s every 15 min; accelerometer was sampled at 10 Hz for 10 s every 30 s. Sampling of GPS creates data gaps by design, and there may be additional gaps due to poor GPS reception (e.g., inside buildings). Gaps must be imputed for valid inference before summary statistics are computed; we used our previously-developed method based on sparse online Gaussian processes to impute missing trajectories (Liu and Onnela, 2021; Barnett and Onnela, 2020), implemented in Forest. GPS data was de-identified to protect privacy by adding random offsets to latitude and longitude coordinates on the phone prior to data upload to the server (Onnela et al., 2021). Based on review of prior digital phenotyping literature and to reduce overlap among features, we pre-selected the following daily summary statistics: (1) home time (total hours at home [GPS]); (2) distance traveled (total kilometers traveled [GPS]); (3) number of significant places (count of distinct places visited, defined as having spent at least 15 min in a location that is at least 50 m from another location [GPS]); (4) total pause time (total hours not in movement [GPS]); (5) location entropy (entropy of the proportion of time spent at unique significant locations [GPS]); (6) steps (count of total steps taken [accelerometer]) (Straczekiewicz et al., 2023a; Straczekiewicz et al., 2023b).

2.4. Statistical analysis

2.4.1. Handling missingness in EMA

Missing data across smartphone summary statistics usually occurred simultaneously for a given participant, such that if one summary statistic was missing, some or all others were missing for that day. Given the robustness of our models to missing data (Little, 1995), we used the 85.9 % of participant days that had both ≥ 1 EMA entry and complete daily smartphone summary statistics.

2.4.2. Feature engineering

We separated features into between- and within-subjects components, to predict outcomes based on sample averages and how deviations from an individual's mean relates to their emotion states, respectively. Between-subjects features were calculated as average sensor data scores for each participant across the 3-month data collection period, representing participants' general behavioral patterns. Within-subjects features were derived by computing deviations of daily sensor variable scores from each participant's mean, capturing day-to-day variations in scores relative to one's typical pattern. While EMA data were collected for 28 days, we extended smartphone sensor data collection to 3 months given its low-burden nature, for more reliable measurement of typical daily patterns when computing features.

2.4.3. Exploratory data analysis

To evaluate whether outcome variables were sufficiently distinct to justify conducting separate analyses for shame, anxiety, and general negative affect, we examined Spearman's correlations (Siegel, 1988) among the three ordinal outcomes (Hauke and Kossowski, 2011).

For each feature and outcome, we generated a Locally Estimated Scatterplot Smoothing (LOESS) plot, which overlays a local polynomial regression line with confidence regions over the scatter plot. We used LOESS plots to inform when to include higher-order terms in linear models, as incorporating all higher-order terms could lead to overfitting and reduced reliability (Harrell Jr., 2015). As RF can make nonlinear transformations on features and create interaction terms while remaining robust to overfitting (Breiman, 2001), we included all first-order features in RF models.

2.4.4. Model descriptions

We conducted three CLMMs (Christensen, n.d.), to predict next-day EMA-rated maximum shame, anxiety, and general negative emotion states, respectively, from GPS- and accelerometer-derived daily summary statistics. We used CLMM, an extension of the generalized linear mixed model, to accommodate ordinal dependent variables. To avoid the potential issue of afternoon features predicting morning emotions on the same day, we use the previous day's features to predict today's emotion outcomes. We included random intercepts. CLMM was implemented using the package *ordinal* (Christensen, n.d.) in R.

To complement CLMMs, we conducted three RFs (Breiman, 2001) to predict next-day maximum shame, anxiety, and general negative emotion states, using all between- and within-person features. RF can generate covariate importance scores and model nonlinear relationships and complex interactions among variables. RFs were implemented using the package *randomforest* (Liaw and Wiener, 2002) in R with standard settings: 500 decision trees per model, allowing each tree to fully develop its prediction pattern, and randomly selecting two variables to consider at each decision point. These settings help the model maintain both accuracy and generalizability.

2.4.5. Variable importance

For each model, we examined variable importance. In CLMM, variable importance is evaluated based on coefficient magnitudes (z-values) and statistical significance. In RF, Gini impurity is used to create decision trees within the forest. It measures how each feature split helps segregate classes within the dataset, with values ranging from 0 to 1;

zero signifies that all items in a node belong to a single class (i.e., perfect purity), and higher values indicate a mix of classes. A higher decrease in gini impurity (i.e., mean decrease gini) demonstrates that a feature significantly reduces impurity, showing greater importance by effectively categorizing mixed classes into more uniform subsets.

2.4.6. Prediction

Predictive performance provides a sense of how useful models might be in future just-in-time interventions. We evaluated predictive performance of the six models (i.e., CLMM and RF for three maximum next-day emotions), using an 80–20 data split to train and test models. We split data at a participant level, such that training data came from earlier observations, while test data came from later observations, ensuring models were tested on future unseen data. RF models directly output a predicted label (i.e., outcome value) from the leaf node. In contrast, CLMM yields predicted probabilities for each level of the dependent variable (i.e., 1–5). Instead of selecting the label with the highest probability, we determined the predicted label by sampling from these probabilities. Sampling introduces variability in predictions, but by representing uncertainty it reduces overconfidence and better reflects the probabilistic nature of its predictions (Agresti, 2010).

We evaluated predictive performance using overall prediction accuracy across all five levels of the dependent variables and prediction accuracy for a binary “high” (i.e., 4, 5) vs. “low” (i.e., 1, 2, 3) version of the dependent variables. Overall accuracy is the proportion of correctly predicted scores out of the total number of data points in the test set. It is a strict measure, due to five levels of our outcomes and the requirement that predicted scores exactly match actual scores. Binary prediction accuracy follows the same principles but assesses the model's ability to distinguish between high and low negative emotion states. Binary prediction accuracy is relevant for future just-in-time interventions, as algorithms might push an intervention when high, but not low, levels of negative emotions are predicted.

3. Results

3.1. Sample characteristics

The initial consented and eligible sample included 87 adults; after removing 4 participants who were missing passive smartphone data ($n = 3$) or had extreme outliers for passive smartphone data ($n = 1$), our final sample included 83 adults with BDD ($M_{\text{age}} = 29.29$ years, $SD_{\text{age}} = 8.87$ years). On average, participants had moderate to severe BDD symptoms ($M_{\text{BDD-YBOCS}} = 30.80$, $SD = 3.53$, range = 24–41) and poor insight into BDD beliefs ($M_{\text{BABS}} = 14.57$, $SD = 3.61$, range = 6–24). See Table 1 for demographic and additional clinical characteristics.

3.2. Exploratory data analysis

3.2.1. Correlations among EMA-reported emotions

Pairwise Spearman's rho correlations between EMA-reported maximum emotions were $\rho = 0.500$ (shame, anxiety), $\rho = 0.710$ (shame, general negative affect), and $\rho = 0.701$ (anxiety, general negative affect) (all $ps < 2.2 \times 10^{-16}$). Values indicate that outcomes are moderately to strongly, positively related but not collinear (e.g., $\rho < 0.9$) (Siegel, 1988). We therefore conducted separate analyses for each emotion state, as this provides a more comprehensive and nuanced understanding of the sensor-based features that predict each outcome.

3.2.2. Feature selection in CLMMs

We used LOESS plots (see Appendix A) to inform whether to include higher-order terms in CLMMs. For between-subjects variables, confidence intervals were narrow at locations where plots show curvature, supporting inclusion of second-order terms. For within-subjects variables, relationships are mostly linear with some curvature at extremes. Confidence intervals are wide where curvature appears; therefore, we

include only first-order terms for within-subjects variables.

3.3. Variable importance

See Appendix B for a table of variable importance statistics across models. In CLMMs, between-subjects home time was the only consistently important variable across emotion outcomes. It was the sole significant feature for the model of maximum general negative affect and was one of only two significant features in the model of maximum shame, along with between-subjects distance traveled. The CLMM for maximum anxiety showed a distinct pattern, with several within- and between-subjects features emerging as important, highlighting its differentiation from the other models. Compared to CLMMs, where between-subjects features were salient, in RFs the within-subjects features contributed greatest predictive value. Specifically, within-subjects distance traveled, steps, entropy, and home time emerged as the most critical features in all RF models.

3.4. Prediction

Table 2 presents overall and binary model prediction accuracies. RFs outperformed CLMMs across performance metrics and emotion outcomes. CLMMs demonstrate low overall prediction accuracy, correctly predicting next-day maximum emotion states 10.9–20.2 % of the time. RF models accurately predicted next-day maximum emotion ratings 42.1–50.0 % of the time, exceeding chance prediction, with highest overall accuracy when predicting general negative emotions and shame, compared to anxiety. For binary classification of high emotion states (scores 4–5), which is clinically salient for risk detection, RF models showed moderate performance. Sensitivity values (i.e., ability to correctly identify high emotion states) ranged from 42.1 to 52.6 %, indicating RF models detected approximately half of all high emotion instances. Specificity values (i.e., ability to correctly identify non-high emotion states) were stronger, ranging from 81.9 to 94.1 %, suggesting RF models were effective at ruling out false positives.

4. Discussion

Risk for suicide is notably elevated in BDD (Phillips and Menard, 2006), and shame and anxiety are well-documented risk correlates (Weingarden et al., 2018; Weingarden et al., 2016; Weingarden et al., 2017). Developing unobtrusive and time-sensitive methods for detecting shame, anxiety, and general negative emotion states using smartphone sensor data can enhance monitoring and enable just-in-time interventions, as next steps toward reducing suicide risk. We evaluated models of maximum daily shame, anxiety, and general negative emotion severity in participants with BDD, using smartphone GPS and accelerometer data. Overall, RF models had moderate performance and outperformed chance prediction, whereas linear models had relatively poor performance and limited utility; models more accurately predicted low

levels of negative emotion, compared to high levels.

To our knowledge, this study follows just one pilot study ($N = 10$) in applying smartphone-based digital phenotyping to BDD (Weingarden et al., 2020) and just one study using EMA in a BDD sample (Schulte et al., 2021). Most prior research developing smartphone-based digital phenotypes in psychiatry has modeled broad mood and emotion states. However, if feasible, detecting specific affect states can enable more nuanced just-in-time interventions. To date, the feasibility of detecting specific emotion states has been under-explored, a limitation that this study addresses.

4.1. Variable importance

We evaluated which variables were most important in models. Across CLMMs, between-subjects home time (Koran et al., 2008) was important. This is consistent with prior literature, which has also pointed to GPS-derived time at home for modeling depression, anxiety, and BDD symptom severity (Stamatis et al., 2024; Chow et al., 2017; Weingarden et al., 2020). Future researchers should collect GPS sensor data and estimate time spent at home, to include in models.

In CLMMs, many more features were associated with anxiety, compared to shame or general negative affect. Specifically, between-person home time and total pause time, and within-person number of significant places and total pause time, were related to anxiety, but not to shame or general negative affect. These features may tap into avoidance, a behavioral manifestation of anxiety. It is possible that we may have captured additional variance in shame had we included call and text log data - passive data streams which likely capture interpersonal dynamics like social withdrawal that reflect behavioral manifestations of shame.

In contrast to CLMMs, which favored between-subjects predictors, within-subjects features were most salient across RFs. RFs underscore the value of collecting accelerometer data to derive within-subjects steps and GPS data to derive within-subjects distance traveled, entropy, and home time in future work. The salience of within-subjects features in RF models, but not CLMMs, suggests that within-subjects sensor features may have nonlinear relationships and complex interactions when predicting negative emotions that RF, but not CLMM, can model. This underscores benefits of nonlinear models in sensor-based digital phenotyping.

4.2. Prediction

The value of using machine learning is further corroborated by comparisons of CLMM and RF predictive performance. As hypothesized and consistent with prior work (Ren et al., 2023), RF outperformed CLMM regardless of outcome. While linear models enhance interpretability by allowing us to understand how each predictor relates to outcomes, for use in just-in-time interventions it likely makes sense to prioritize predictive performance above interpretability, especially when performance is so discrepant across linear vs. nonlinear approaches.

Overall model accuracy for RF was moderate and exceeded chance. Roughly half the time, models accurately predicted the exact ordinal emotion severity rating (1–5) (a conservative metric for accuracy), given passively collected features alone.

In contrast to hypotheses that prediction accuracy would be highest for general emotions, RF models' overall prediction of general negative emotion severity and shame severity were both more accurate than prediction of anxiety severity. While most prior digital phenotyping work has focused on general emotion states, this study was unique in its attempt to predict specific affect states. It is encouraging that even complex social emotions like shame can be modeled with moderately strong accuracy in this proof-of-concept study. There are benefits to predicting specific emotion states. Whereas prediction of general negative emotions may inform delivery of broad cognitive-behavioral

Table 2
Model performance metrics for predicting negative emotion states.

Model	Outcome	Overall accuracy	Sensitivity (high emotion)	Specificity (high emotion)
CLMM ^a	Anxiety	0.109	0.445	0.584
	General negative affect	0.116	0.368	0.724
	Shame	0.202	0.288	0.712
RF ^b	Anxiety	0.421	0.526	0.819
	General negative affect	0.50	0.421	0.941
	Shame	0.491	0.452	0.911

Note: Overall accuracy represents exact match across all five rating levels.

^a CLMM = cumulative link mixed model.

^b RF = random forest.

strategies, just-in-time detection of anxiety may *specifically* inform prompts to complete exposure exercises, a frontline strategy for anxiety; detection of shame may trigger delivery of self-compassion exercises. Delivering affect-matched skills may provide added intervention effect. Future researchers should continue to refine passive measurement of specific emotions.

Because one potential use of models is risk prediction (i.e., predicting elevated negative emotion states that may predispose someone to suicidal thinking), it is especially important that models demonstrate predictive accuracy at high levels (i.e., scores of 4, 5) of negative affect, which are likely to be used as a threshold for delivering just-in-time interventions. Thus, we evaluated binary predictive performance for high emotion states. Again, RFs had stronger performance than CLMMs. At high levels of negative emotion, RF models demonstrated moderate performance, with sensitivity ranging from 42.1 to 52.6 % and specificity ranging from 81.9 to 91.4 %. Sensitivity represents the model's ability to correctly identify true high emotion states, while specificity indicates the model's ability to correctly reject non-high emotion states (i.e., reduce false positives). While our models showed a strong ability to minimize unnecessary alerts – important for managing patient and provider burden with an alert system – they only correctly identified high-risk events up to half the time and would thus fail to provide potentially necessary interventions relatively often. Researchers must focus next on how to improve prediction for high negative emotion states, prior to deploying in just-in-time interventions. To optimize the balance of sensitivity and specificity at high levels of negative emotion, future research could develop adaptive algorithms that dynamically adjust sensitivity based on real-time feedback and changing patient conditions. Moreover, researchers could include select low-burden variables to complement passive data such as demographic features, which would not impede feasibility and may meaningfully strengthen predictive performance.

4.3. Limitations and next steps

This study employed EMA and digital phenotyping in BDD for one of the first times. With 83 participants, our sample was large compared to extant BDD research but relatively small for digital phenotyping. Due to the burden of collecting EMA 3 times daily, we collected emotion ratings for 28 days; future studies could collect EMA-rated outcomes fewer times per day over more days, to increase days with both sensor and EMA data and, subsequently, power.

In addition to incorporating demographic features (noted above), there may be value in complementing smartphone sensor data with physiological data from wearables (e.g., heart rate variability, skin conductance), which may capture emotion correlates like stress and arousal. That said, while wearable data may strengthen model performance, it limits model generalizability, because it requires individuals to purchase and consistently use an additional device. A strength of the present study is that models rely only on one's personal smartphone, making models applicable to the vast majority.

While RF models assumed data were independently and identically distributed, this assumption may lead to overfitting when there is within-person correlation in longitudinal data. As a result, RF models might show overly optimistic performance on training data, but generalize less well to new data.

Finally, to reduce overfitting, we pre-selected six smartphone summary statistics to explore in models, from a larger set of available summary statistics. We made selections by drawing on past research, conceptualization of which metrics likely map onto emotion states, seeking to reduce collinearity in features, and exploratory data analysis. However, it may be valuable to investigate the broader set of available smartphone summary statistics in future work.

Altogether, as next steps, models might be strengthened with larger samples, outcome data collected over longer time frames, and addition of demographic, call and text, or wearable-based physiological data, to

supplement models' predictive ability.

Ultimately, we wish to test how models such as these can be harnessed for just-in-time intervention. People with BDD are often severely ill – even homebound due to anxiety or shame – and may benefit tremendously from timely, affect-matched support. Risk prediction algorithms can be integrated into care in several ways. First, people can use a risk prediction program on their smartphone for self-help use. When the algorithm predicts an uptick in shame or anxiety, a mobile app would push an alert to the user to (1) build emotion awareness (e.g., “you may be feeling anxious right now”) and (2) suggest appropriate skills (e.g., cognitive restructuring, exposure, ritual prevention). The goal is to manage high-risk affect states *before* the user escalates to worsened downstream outcomes such as suicidal thinking. However, if the model detects continued high negative emotion, it could next offer tools to support safety, such as National Suicide Hotline and Crisis Textline information and safety planning support. Beyond self-help use, risk prediction algorithms can be integrated into smartphone-delivered CBT for BDD (Wilhelm et al., 2022) or traditional face-to-face therapy, where predictive algorithms not only communicate with the user but also with their clinician, with user consent.

4.4. Conclusions

We built linear and nonlinear models of next-day maximum negative emotion severity in participants with BDD, using smartphone GPS and accelerometer data. Predicting elevated negative affect is a first step in developing just-in-time interventions that target these emotion states, with the aim of reducing risk for associated outcomes like suicide.

RF models had moderate performance. Time at home, entropy, distance traveled, and steps were the most salient features. Replication and continued work to enhance model accuracy, especially at high levels of negative emotions, are critical before models can be tested in real-world clinical contexts. Results provide a first indication that passively-collected smartphone sensor data can predict clinically relevant general and specific emotions in BDD.

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Declaration of competing interest

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